

CS: assignment 2.

(Q.1) Explain Cyberterrorism, its Meaning and objective and how it is different from cyber crime.

- Cyberterrorism refers to the use of Computer networks or digital technologies to carry out attack intended to cause serious disruption, fear, or damage to a government.
- Objective of Cyberterrorism:
- i. Creating fear and panic among the public
 - ii. Disrupting critical infrastructure such as electricity, transportation and communication
 - iii. Targeting government system and public
 - iv. Causing economic damage to a country or organization.

Cyberterrorism

Cyber Crime

Can have national security implications

Generally treated as a Criminal & Cyberterrorism

usually motivated by political ideological or similar objective.

usually motivated by financial gain, personal benefit, or other criminal.

May target government critical infrastructure or public system

Commonly targets individuals, businesses, or financial system

2. What are the Common targets of Cyber attacks? Explain with examples and their impact on National security and the Economy.

1. Government systems:
websites, databases and internal networks
Can be targeted to steal sensitive information or disrupt public services.

Eg: An attacker may compromise a government database containing

2. Critical Infrastructure

Critical infrastructure includes systems such as: Water supply, system, Transportation

Eg: An attack on an electricity management system could disrupt power distribution

3. Financial institutions

Bank, Stock exchanges, payment system and financial companies are also common targets because they contain valuable financial information.

4. Healthcare organizations

Hospitals and healthcare systems store sensitive personal information and depend heavily on computer systems.

Example: A ransomware attack could prevent hospital staff from accessing important computer systems.

3. What are the impacts of cyber attacks on National security and the Economy?

→ Here

1. Government

2. Infrastructure

3. Financial

4. Healthcare

3. What measures Can Government & Organisations take to prevent and Respond to Cyberterrorism ? what is the role of Cyber Security ?

→ Measures Government Can Take

1. Develop Cybersecurity policies:
Government should establish optional cybersecurity policies, standard and legal framework for protecting critical infrastructure.
2. protect Critical Infrastructure.
Important system such as power, transportation, healthcare, and communications network should have strong security controls.
3. Establish Incident Response Teams.
government should maintain specialized teams that can detect, investigate and respond to major cyber incident.
4. Information Sharing
government agencies should share information about emerging threats, vulnerabilities and attacks with relevant organisations.
5. International Co-operation
Cyber attacks can originate from different countries, governments can cooperate internationally to investigate threats and strengthen cyber security.

Measures organisation Can take.

1. Strong Access Control
Only authorized employees should be allowed to access sensitive systems.

2. Multi-Factor Authentication
MFA provides an additional security layer beyond password.

3. Regular Software updates
Operating systems, applications, and security tools should be regularly patched to reduce known vulnerabilities.